

# Altasker Conceptual ER — Unified Draw Plan

(Merges: original 10-phase plan + Amendment Series A + Amendment Series B. B4 and B5 are absorbed — cardinality was already correct and the signal→claim relationship is covered in Phase 2.)

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## Entities to Draw — 51 Total

**Role & auth:** users · client\_profiles · expert\_profiles · tech\_team\_profiles

**Expert capability:** expert\_domain\_depths · expert\_seam\_claims · expert\_stack\_tags · portfolio\_submissions · scenario\_assessments · scenario\_responses · expert\_seam\_outcome\_signals

**Finance:** wallets · wallet\_transactions · virtual\_accounts · user\_subscriptions · withdrawal\_requests · platform\_settings

**Org:** organizations · organization\_members

**Elicitation & projects:** elicitation\_sessions · projects · capability\_footprints · artifact\_a · artifact\_b

**Services:** services

**Engagements & bids:** engagements · spec\_clarifications · capability\_bids · bid\_versions · bid\_revision\_requests · price\_negotiations · bid\_conflict\_overrides

**Milestones:** milestones · acceptance\_criteria · revision\_requests · milestone\_dod\_items · milestone\_dod\_items · milestone\_sprints · sprint\_status\_updates · milestone\_submissions · paygated\_documents · escrow\_accounts · addon\_phase\_requests

**Sprint & DoD comments:** sprint\_comments · dod\_item\_comments

**Disputes:** disputes · dispute\_resolution\_reports

**Messaging & reads:** messages · message\_reads

**Reviews & signals:** reviews · outcome\_signals

**Platform:** platform\_decisions

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## Phase 0 — Elicitation Engine

Draw **elicitation\_sessions** positioned between **users** and **projects** on the canvas. It will chain into **projects** in Phase 5.

- **users** ❖ **initiates** → **elicitation\_sessions (1:N)** — CEO initiates sessions; one session per Path A attempt
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## Phase 1 — Users & Role Profile Subtypes

Draw **users** as the central anchor entity for the entire diagram.

- **users** ⬢ has → **client\_profiles** (1:1)
- **users** ⬢ has → **expert\_profiles** (1:1)
- **users** ⬢ has → **tech\_team\_profiles** (1:1)
- **users** ⬢ invited → **tech\_team\_profiles** (1:N) — via **linked\_client\_id**; CEO who issued the handoff link

(The fifth relationship **projects** → **scopes** → **tech\_team\_profiles** is drawn in Phase 5 when **projects** exists.)

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## Phase 2 — Expert Capability Map

All branch from **expert\_profiles**. Draw **expert\_seam\_outcome\_signals** here too — it lives in the expert cluster but connects back to **outcome\_signals** in Phase 10.

- **expert\_profiles** ⬢ has → **expert\_domain\_depths** (1:N)
- **expert\_profiles** ⬢ holds → **expert\_seam\_claims** (1:N)
- **expert\_profiles** ⬢ has → **expert\_stack\_tags** (1:N)
- **expert\_profiles** ⬢ submits → **portfolio\_submissions** (1:N)
- **expert\_profiles** ⬢ submits → **scenario\_responses** (1:N)
- **scenario\_assessments** ⬢ receives → **scenario\_responses** (1:N)
- **expert\_seam\_claims** ⬢ updated by → **expert\_seam\_outcome\_signals** (1:N) — via **seam\_claim\_id**; Tier 4 accumulation write-back
- **expert\_profiles** ⬢ has → **expert\_seam\_outcome\_signals** (1:N)

(Two more lines into **expert\_seam\_outcome\_signals** — from **engagements** and **outcome\_signals** — are drawn in Phases 7 and 10 respectively when those entities exist.)

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## Phase 3 — Wallet, Finance & Subscriptions

Draw **wallets** beside **users**. Draw **platform\_settings** as a singleton near the wallet cluster with a note: "one row seeded at deploy".

- **users** ⬢ owns → **wallets** (1:1)
- **wallets** ⬢ records → **wallet\_transactions** (1:N)
- **users** ⬢ has → **user\_subscriptions** (1:N) — 1:N because a dual-role user holds two rows, one per **role\_type**
- **expert\_profiles** ⬢ creates → **withdrawal\_requests** (1:N)
- **platform\_settings** ⬢ references → **wallets** (1:1) — the seeded platform fee wallet row

(The **virtual\_accounts** relationship is drawn in Phase 9 when **milestones** exists.)

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## Phase 4 — Organizations

- `users` ✦ `owns` → `organizations` (1:N) — via `owner_user_id`
- `organizations` ✦ `has` → `organization_members` (1:N)
- `users` ✦ `joins` → `organization_members` (1:N)

## Phase 5 — Projects & Dual Artifacts

Draw `projects` centrally — it is the second major hub of the diagram.

- `users` ✦ `creates` → `projects` (1:N) — via `client_id`; CEO only
- `elicitation_sessions` ✦ `produces` → `projects` (1:1) — Stage 5 synthesis output; primary Path A route
- `projects` ✦ `scopes` → `tech_team_profiles` (1:N) — via `linked_project_id`; completes the handoff link chain from Phase 1
- `projects` ✦ `has` → `capability_footprints` (1:1)
- `capability_footprints` ✦ `drives` → `artifact_a` (1:1) — synthesis engine constructs Artifact A from the footprint
- `projects` ✦ `has` → `artifact_b` (1:1) — data parent is always Project; access is gated by engagement state

(The `engagements` → `gates access to` → `artifact_b` line is drawn in Phase 7 when `engagements` exists.)

## Phase 6 — Services (Path B)

Draw `services` beside `expert_profiles`.

- `expert_profiles` ✦ `creates` → `services` (1:N)

(The `services` → `generates` → `engagements` line is drawn in Phase 7.)

## Phase 7 — Engagements

Draw `engagements` as the second major hub. Five entities feed into it; two access gate lines close back to Phase 5.

### Inbound feeds:

- `projects` ✦ `has` → `engagements` (1:N) — Path A; `project_id` nullable for Path B
- `expert_profiles` ✦ `joins` → `engagements` (1:N)
- `organizations` ✦ `participates in` → `engagements` (1:N) — via `organization_id`; nullable
- `services` ✦ `generates` → `engagements` (1:N) — Path B only; `service_id` nullable for Path A

### Artifact B access gate (closes Phase 5):

- `engagements` ✦ `gates access to` → `artifact_b` (N:1) — annotate: "`state` ≥ `CONNECTED` + `NDA accepted`; `CEO permanently excluded`"

### Surface A — Spec clarifications:

- `artifact_a` ✧ `receives` → `spec_clarifications` (1:N) — pre-bid questions filed against Artifact A
- `expert_profiles` ✧ `files` → `spec_clarifications` (1:N) — who asked
- `users` ✧ `responds to` → `spec_clarifications` (1:N) — via `responded_by`; TECH\_TEAM on Tier 2+ projects, CEO on Tier 1

### Expert seam outcome signals (closes Phase 2):

- `engagements` ✧ `has` → `expert_seam_outcome_signals` (1:N)

### Escrow Path B (SERVICE\_PURCHASE only — no milestone exists at IPN time):

- `engagements` ✧ `holds escrow for` → `escrow_accounts` (1:1) — annotate: `"SERVICE_PURCHASE only"`

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## Phase 8 — Non-Linear Bid System (Surfaces A–D)

Draw `capability_bids` branching from `engagements`.

### Core bid:

- `engagements` ✧ `has` → `capability_bids` (1:1) — one bid per engagement; absent for SERVICE\_PURCHASE

### Surface B — Revision chain:

- `capability_bids` ✧ `has` → `bid_versions` (1:N) — immutable snapshot on every REVISED transition
- `capability_bids` ✧ `receives` → `bid_revision_requests` (1:N) — TECH\_TEAM flags one component
- `bid_revision_requests` ✧ `produces` → `bid_versions` (1:1) — each request produces one version snapshot
- `users` ✧ `requests` → `bid_revision_requests` (1:N) — via `requested_by`
- `users` ✧ `authors` → `bid_versions` (1:N) — via `reviser_id`

### Surface C — Price negotiation:

- `capability_bids` ✧ `has` → `price_negotiations` (1:N) — max 2 rounds; bounded by CHECK constraint

### Surface D — CEO conflict override:

- `capability_bids` ✧ `has` → `bid_conflict_overrides` (1:N)
  - `users` ✧ `files override` → `bid_conflict_overrides` (1:N) — via `ceo_user_id`; CEO who bypassed TECH disapproval
  - `bid_conflict_overrides` ✧ `logged in` → `platform_decisions` (1:1) — Platform Integrity Monitor feed
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## Phase 9 — Milestone System (3 Layers) + Add-On + Disputes

Draw **milestones** branching from **engagements**. This is the most complex phase — work layer by layer.

### Engagement → Milestones:

- **engagements** ✧ **has** → **milestones (1:N)**
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### Layer 1 — Acceptance Criteria (payment contract):

- **milestones** ✧ **has** → **acceptance\_criteria (1:N)** — all required criteria must have **verified\_at** set before APPROVED
  - **acceptance\_criteria** ✧ **has** → **revision\_requests (1:N)**
  - **users** ✧ **files** → **revision\_requests (1:N)** — via **requested\_by**
  - **acceptance\_criteria** ✧ **evaluation logged in** → **platform\_decisions (1:1)** — LLM quality gate on criterion save; subjective language flagged and returned
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### Layer 2 — DoD Checklist (gates SUBMITTED):

- **milestones** ✧ **has** → **milestone\_dod\_items (1:N)** — all required items must reach COMPLETED before submission is permitted
  - **milestone\_dod\_items** ✧ **maps to** → **acceptance\_criteria (N:1)** — via **maps\_to\_criterion\_id**; nullable; optional upward link
  - **milestone\_dod\_items** ✧ **has** → **dod\_item\_comments (1:N)** — TECH\_TEAM per-DoD-item comment thread; separate from main channel
  - **users** ✧ **writes** → **dod\_item\_comments (1:N)** — TECH\_TEAM only at route level
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### Layer 3 — Sprint Plan & Status:

- **milestones** ✧ **has** → **milestone\_sprints (1:N)** — temporal decomposition; no payment consequence
  - **milestone\_sprints** ✧ **has** → **sprint\_comments (1:N)** — TECH\_TEAM per-sprint comment thread
  - **users** ✧ **writes** → **sprint\_comments (1:N)** — TECH\_TEAM only at route level
  - **milestone\_sprints** ✧ **receives** → **sprint\_status\_updates (1:N)** — via **milestone\_sprint\_id**; direct FK replacing implicit **sprint\_number** link
  - **users** ✧ **submits** → **sprint\_status\_updates (1:N)** — EXPERT only
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### Add-On Phase (F8 Scope Evolution):

- **sprint\_status\_updates** ✧ **triggers** → **addon\_phase\_requests (1:1)** — annotate: "SCOPE\_EVOLUTION blocker only"; UNIQUE constraint on trigger side
  - **engagements** ✧ **has** → **addon\_phase\_requests (1:N)**
  - **users** ✧ **submits** → **addon\_phase\_requests (1:N)** — via **submitted\_by**; EXPERT files the brief
  - **users** ✧ **tech approves** → **addon\_phase\_requests (1:N)** — via **tech\_approved\_by**; independent approval
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- `users` ❖ `ceo approves` → `addon_phase_requests` (1:N) — via `ceo_approved_by`; independent approval
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### Deliverables & Pay-gated Documents:

- `milestones` ❖ `has` → `milestone_submissions` (1:N)
  - `expert_profiles` ❖ `submits` → `milestone_submissions` (1:N) — via `expert_id`
  - `milestones` ❖ `triggers release of` → `paygated_documents` (1:N) — `release_state` flips STAGED→RELEASED when milestone IPN fires; TECH\_TEAM-only read
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### Escrow (closes Phase 7 Path B line):

- `milestones` ❖ `held in` → `escrow_accounts` (1:1) — PROJECT\_BASED path; one escrow per milestone
  - `wallets` ❖ `funds` → `escrow_accounts` (1:N) — via `client_wallet_id`
  - `wallets` ❖ `receives from` → `escrow_accounts` (1:N) — via `expert_wallet_id`
  - `milestones` ❖ `allocates` → `virtual_accounts` (1:N) — per-milestone VA for VietQR payment
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### Disputes:

- `engagements` ❖ `has` → `disputes` (1:N)
  - `milestones` ❖ `has` → `disputes` (1:N)
  - `acceptance_criteria` ❖ `subject of` → `disputes` (1:N) — disputes always filed against a specific criterion
  - `users` ❖ `files` → `disputes` (1:N) — CEO, TECH\_TEAM, or EXPERT can all file
  - `disputes` ❖ `freezes` → `escrow_accounts` (1:1) — `escrow.status` → FROZEN immediately on filing
  - `disputes` ❖ `has` → `dispute_resolution_reports` (1:N)
  - `users` ❖ `files` → `dispute_resolution_reports` (1:N) — the party who felt the Layer 3 resolution was unfair
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## Phase 10 — Messaging, Reviews & Platform Decisions

### Messaging:

- `engagements` ❖ `has` → `messages` (1:N) — direct `engagement_id` FK; no thread table
- `users` ❖ `sends` → `messages` (1:N) — via `sender_id`; any of the three transactional roles
- `messages` ❖ `has` → `message_reads` (1:N) — per-user read record
- `users` ❖ `generates` → `message_reads` (1:N) — one row per user × message pair; enables per-role unread badge

### Reviews & Outcome Signals:

- `engagements` ❖ `has` → `reviews` (1:N)
  - `users` ❖ `writes` → `reviews` (1:N) — via `reviewer_id`
  - `users` ❖ `reviewed as` → `reviews` (1:N) — via `target_id`
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- **engagements** ✦ **has** → **outcome\_signals** (1:N)
- **reviews** ✦ **generates** → **outcome\_signals** (1:1) — TECH\_TEAM structured review produces one aggregate signal record

### Closes Phase 2 — Outcome Signals → Expert Seam Outcome Signals:

- **outcome\_signals** ✦ **generates** → **expert\_seam\_outcome\_signals** (1:N) — NestJS extracts per-seam rows from the JSONB aggregate after each review save; these rows power Tier 4 accumulation and the matching hard gate

**Platform Decisions** — passive audit log, written by FastAPI only. Already has two incoming lines from Phase 8 (**bid\_conflict\_overrides**) and Phase 9 (**acceptance\_criteria**). No further relationships — **entity\_id** and **entity\_type** are polymorphic by design.

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## Final Count

| Phase                              | Entities  | Relationships |
|------------------------------------|-----------|---------------|
| 0 — Elicitation Engine             | 1         | 1             |
| 1 — Users & Role Subtypes          | 4         | 4             |
| 2 — Expert Capability              | 7         | 8             |
| 3 — Wallet & Finance               | 6         | 5             |
| 4 — Organizations                  | 2         | 3             |
| 5 — Projects & Artifacts           | 5         | 6             |
| 6 — Services                       | 1         | 1             |
| 7 — Engagements                    | 1         | 8             |
| 8 — Bid System                     | 7         | 10            |
| 9 — Milestones, Add-On, Disputes   | 13        | 35            |
| 10 — Messaging, Reviews, Decisions | 4         | 10            |
| <b>Total</b>                       | <b>51</b> | <b>91</b>     |